

CHATEAUDUN AB, France

WMO: 071400

Lat: 48.0614N

Lon: 1.3764E

Elev: 132

StdP: 99.75

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.3	-3.5	-8.1	1.9	-3.6	-6.1	2.3	-2.3	13.5	9.5	12.3	9.3	2.4	40	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	31.3	19.9	29.2	19.1	27.1	18.5	21.1	28.6	20.1	26.9	19.2	25.3	3.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	13.9	23.3	17.8	13.0	22.4	17.0	12.3	21.4	61.7	28.7	58.1	26.9	55.1	25.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.6	8.6	DB	-8.5	35.4	3.3	2.2	-10.9	37.0	-12.8	38.2	-14.7	39.5	-17.1	41.0	(4)
(5)			WB	-8.7	23.1	3.2	1.1	-11.0	23.9	-12.8	24.5	-14.6	25.1	-16.9	25.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	4.2	4.7	7.4	10.1	13.7	17.1	19.3	18.7	16.0	12.1	7.4	4.6	(6)	
	DBStd	6.40	3.81	3.84	3.40	3.17	3.23	3.34	3.26	2.97	2.91	3.43	3.47	3.80	(7)	
	HDD10.0	748	181	151	94	37	6	0	0	0	0	19	91	171	(8)	
	HDD18.3	2702	438	383	338	247	148	62	27	30	82	193	328	426	(9)	
	CDD10.0	1229	2	2	14	39	120	214	287	271	180	84	13	3	(10)	
	CDD18.3	139	0	0	0	0	3	26	56	43	12	1	0	0	(11)	
	CDH23.3	1643	0	0	0	11	57	306	641	475	142	11	0	0	(12)	
(13)	CDH26.7	536	0	0	0	1	6	97	232	166	34	1	0	0	(13)	
(14)	Wind	WSAvg	4.1	4.5	4.7	4.5	4.1	3.8	3.7	3.8	3.6	3.6	3.8	4.2	4.5	(14)
(15) Precipitation	PrecAvg	631	55	46	42	49	56	52	55	45	45	66	58	62	(15)	
	PrecMax	857	115	102	132	121	115	104	170	102	119	147	107	138	(16)	
	PrecMin	455	7	4	7	8	11	8	16	8	4	20	18	13	(17)	
	PrecStd	110	26	25	29	30	27	26	33	28	28	32	24	28	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	15.7	20.8	24.9	27.5	33.2	34.6	33.8	30.0	24.8	17.1	13.8	(19)	
		MCWB	11.9	10.5	12.2	15.6	19.2	21.7	20.5	20.4	18.3	17.4	14.7	12.5	(20)	
	2%	DB	12.1	12.9	17.8	21.6	24.9	29.4	31.4	30.7	27.2	21.5	15.3	12.6	(21)	
		MCWB	11.0	10.1	11.7	14.1	17.5	19.9	19.8	19.5	17.9	16.4	13.8	11.5	(22)	
	5%	DB	10.9	11.5	15.3	18.9	22.8	26.8	29.2	28.3	24.6	19.1	13.9	11.6	(23)	
		MCWB	9.9	9.6	10.9	12.9	16.5	18.8	19.1	18.8	17.1	15.3	12.4	10.5	(24)	
	10%	DB	9.7	10.3	13.3	16.5	20.7	24.4	27.0	25.8	22.4	17.3	12.6	10.5	(25)	
		MCWB	8.7	8.8	10.2	11.6	15.3	17.7	18.5	17.8	16.2	14.5	11.4	9.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.4	11.7	13.6	16.6	20.4	22.9	22.1	21.9	20.2	18.5	15.6	12.8	(27)	
		MCDB	12.9	13.5	17.5	23.6	26.0	29.3	31.6	31.0	25.9	22.3	16.4	13.4	(28)	
	2%	WB	11.2	10.7	12.5	14.7	18.4	21.2	20.9	20.5	19.0	17.0	14.0	11.7	(29)	
		MCDB	11.9	12.1	15.7	20.2	23.5	27.6	29.0	27.7	24.8	19.8	15.2	12.4	(30)	
	5%	WB	10.0	9.9	11.6	13.4	17.0	19.6	19.9	19.5	17.9	16.0	12.7	10.7	(31)	
		MCDB	10.8	11.2	14.4	17.9	21.6	25.3	27.2	25.9	23.2	18.6	13.8	11.5	(32)	
	10%	WB	8.8	8.9	10.6	12.1	15.7	18.4	19.0	18.6	17.0	15.0	11.5	9.5	(33)	
		MCDB	9.5	10.3	13.2	15.8	19.7	23.5	25.4	24.3	21.3	17.0	12.5	10.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	7.4	9.5	10.8	10.9	11.4	12.1	12.1	11.6	9.0	6.6	5.8	(35)	
		MCDBR	6.2	9.5	13.2	14.6	14.3	15.3	16.1	16.4	15.3	11.4	7.6	6.4	(36)	
	5% WB	MCWBR	5.4	7.0	7.8	8.1	7.7	7.1	6.1	6.7	7.2	6.9	6.0	5.4	(37)	
		MCWBR	5.9	8.2	10.9	13.2	13.0	14.0	14.2	13.9	12.9	10.2	7.4	6.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.321	0.341	0.370	0.398	0.402	0.409	0.405	0.394	0.367	0.360	0.333	0.317	(40)	
		taud	2.435	2.382	2.297	2.224	2.254	2.270	2.275	2.325	2.385	2.414	2.439	2.409	(41)	
	Ebn at Noon		749	812	840	851	861	854	852	846	836	779	726	700	(42)	
		Edh at Noon	67	86	110	130	131	130	128	116	100	83	66	61	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.98	1.79	2.97	4.41	5.19	5.81	5.56	4.87	3.78	2.23	1.20	0.84	(44)	
	RadStd		0.10	0.24	0.38	0.51	0.46	0.57	0.39	0.46	0.28	0.19	0.12	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page